
Arduino Robotics & Electronics — Lesson Plan

Traffic Light Simulation

Age Group: 6–9 years | **Duration:** 90 minutes | **Difficulty:** Beginner

Organization: TinkerWithMe, Nairobi

1. Session Overview

In this hands-on session, young learners will build a working **traffic light system** using an Arduino Uno, three LEDs, and a breadboard. Students will connect real electronic components and write their very first lines of code to control the timing sequence of red, yellow, and green lights — just like the traffic lights they see on the streets of Nairobi.

This project matters because it introduces children to the foundational concepts of *circuits*, *programming logic*, and *cause-and-effect sequencing* in a context they already understand from everyday life. By the end of the session, students will have built and coded a real device that lights up in a repeating pattern, giving them a strong sense of accomplishment and curiosity for further STEM exploration.

2. Learning Objectives

By the end of this session, students will be able to:

- **Remember:** Identify the three colors of a traffic light and recall their real-world meaning (stop, slow down, go).

- **Understand:** Explain in simple words how an LED lights up when connected correctly to power.
- **Apply:** Build a simple circuit on a breadboard by correctly placing LEDs, resistors, and jumper wires with guidance.
- **Apply:** Upload and run a basic Arduino sketch that turns LEDs on and off in a timed sequence.
- **Analyze:** Identify a simple wiring or code mistake (e.g., LED not lighting) with support from a teacher or peer.

3. Materials & Setup

Recommended group size: **2 students per Arduino kit**, with a teacher or helper supporting every 2 groups.

ITEM	SPECIFICATION	QUANTITY (PER GROUP)
Arduino Uno Board	Uno R3 or compatible, USB powered	1
USB Cable	Type A to Type B	1
Breadboard	Half-size, 400 tie points	1
LED	5mm Red	1
LED	5mm Yellow	1
LED	5mm Green	1
Resistor	220 ohm, 1/4 watt	3
Jumper Wires	Male-to-male, assorted lengths	8
Laptop/Computer	With Arduino IDE pre-installed	1
Printed Traffic Light Diagram	A4 laminated wiring reference sheet	1
Name/Group Labels	Stickers or tape	2

Room Setup Checklist

- Arrange tables in pods of 2–4 students for easy teacher access.
- Pre-load the Arduino IDE and confirm each laptop recognizes the Arduino board (drivers installed) before the session begins.
- Pre-cut jumper wires if working with very young learners to save time.

- Place a laminated wiring diagram at each station as a visual reference.
- Have spare LEDs and resistors on hand — young learners often bend leads or insert components backwards.

4. Minute-by-Minute Schedule

TIME	ACTIVITY	DETAILS
0:00 – 0:05	Welcome & Hook	Greet students. Ask: "Who has seen a traffic light? What do the colors mean?" Build excitement for building a real one.
0:05 – 0:15	Unplugged Discussion & Demo	Show a pre-built working traffic light model. Explain LEDs, resistors, and breadboards using simple analogies (e.g., "resistor is like a narrow pipe that controls water flow").
0:15 – 0:20	Materials Distribution & Safety Briefing	Hand out kits. Explain safe handling of components (no bending wires forcefully, careful with LED legs).
0:20 – 0:45	Guided Circuit Building	Step-by-step wiring of LEDs, resistors, and jumper wires on breadboard, following teacher demonstration one step at a time.
0:45 – 0:50	Stretch Break — "Red Light, Green Light" Game	Physical movement game reinforcing traffic light meaning while giving hands a rest.
0:50 – 1:10	Coding Time	Guided typing of the Arduino sketch as a class, explaining each line simply. Upload and run first test.
1:10 – 1:20	Debug & Customize	Students test their circuit, fix issues with help, and try changing delay times to make lights faster/slower.
1:20 – 1:25	Assessment & Mini Showcase	Each group demonstrates their working traffic light to the class.
1:25 – 1:30	Clean Up & Follow-Up Challenge	Pack away components carefully. Introduce take-home challenge.

5. Project Breakdown: Traffic Light Simulation

Step-by-Step Instructions

- **Step 1:** Place the breadboard in front of you with the Arduino board beside it.
- **Step 2:** Insert the red LED into the breadboard — long leg (positive) in one row, short leg (negative) in another row.
- **Step 3:** Connect a 220-ohm resistor from the LED's short leg row to the breadboard's ground rail (marked with a blue/black line).

- **Step 4:** Repeat Steps 2–3 for the yellow and green LEDs, placing them side by side.
- **Step 5:** Use jumper wires to connect the long leg (positive) of each LED to Arduino digital pins: Red → Pin 8, Yellow → Pin 9, Green → Pin 10.
- **Step 6:** Connect the breadboard's ground rail to the Arduino's **GND** pin using a jumper wire.
- **Step 7:** Double-check all connections against the laminated wiring diagram with your teacher or helper.
- **Step 8:** Connect the Arduino to the laptop using the USB cable.
- **Step 9:** Open the Arduino IDE, type (or copy) the sketch below, and click *Upload*.
- **Step 10:** Watch your traffic light come to life! Adjust delay times to experiment.

Full Working Code

```
// -----
// TinkerWithMe: Traffic Light Simulation
// For ages 6-9 - simple ON/OFF light sequence
// -----

int redLED = 8;      // Red LED connected to pin 8
int yellowLED = 9;   // Yellow LED connected to pin 9
int greenLED = 10;   // Green LED connected to pin 10

void setup() {
  // Tell the Arduino these pins will send power OUT to the LEDs
  pinMode(redLED, OUTPUT);
  pinMode(yellowLED, OUTPUT);
  pinMode(greenLED, OUTPUT);
}

void loop() {
  // GREEN LIGHT - cars can GO!
  digitalWrite(greenLED, HIGH); // turn green LED ON
  delay(4000);                  // wait 4 seconds
  digitalWrite(greenLED, LOW);  // turn green LED OFF

  // YELLOW LIGHT - get ready to stop!
  digitalWrite(yellowLED, HIGH); // turn yellow LED ON
  delay(1500);                   // wait 1.5 seconds
  digitalWrite(yellowLED, LOW);  // turn yellow LED OFF

  // RED LIGHT - cars must STOP!
  digitalWrite(redLED, HIGH);    // turn red LED ON
  delay(4000);                  // wait 4 seconds
  digitalWrite(redLED, LOW);     // turn red LED OFF
}
```

```
// Then the loop repeats forever!  
}
```

Common Mistakes & How to Fix Them

- **Mistake:** LED does not light up.
Fix: Check that the LED's long leg (positive) is in the correct row connected to the Arduino pin, and the short leg is connected through the resistor to ground.
- **Mistake:** LED inserted backwards.
Fix: Flip the LED around — it will only light up in one direction.
- **Mistake:** No resistor used, LED burns out or gets very hot.
Fix: Always use the 220-ohm resistor in series with each LED.
- **Mistake:** Code uploads but nothing happens.
Fix: Confirm the correct board and COM port are selected in the Arduino IDE menu (Tools > Board / Port).
- **Mistake:** Wires loose or falling out of breadboard.
Fix: Push wires firmly into breadboard holes; ask a helper to check connections.

Success Criteria

- All three LEDs light up in the correct order: Green → Yellow → Red.
- Student can explain, in their own words, what each color means.
- Circuit runs continuously without loose connections.
- Student successfully uploads code with minimal teacher assistance.

6. Assessment

- **Mini Showcase:** Each group demonstrates their working traffic light to the class and explains what each color means.
- **Quick Quiz Game:** Teacher calls out a color; students point to the matching LED on their board and say what it means (stop/slow/go).
- **Observation Checklist:** Teacher walks around during build time, checking each student can identify LED polarity and explain the resistor's purpose in simple terms.
- **Peer Teaching:** Fast-finishing students help a neighboring group troubleshoot, reinforcing their own understanding.

7. Extensions & Challenges

- **Speed Challenge:** Change the delay times to make the lights change faster or slower — can they create a "super-fast" traffic light?
- **Pedestrian Light:** Add a fourth LED to represent a pedestrian "walk" signal that turns on when the light is red.
- **Blink Pattern:** Make the yellow LED blink three times quickly before turning red, mimicking some real-world traffic lights.
- **Storytelling:** Ask advanced students to narrate a short story about a car driving through their traffic light system.

8. Troubleshooting Guide

Frequently Asked Questions

- **Q: My LED is dim or flickering.**
A: Check that the resistor is properly connected and not damaged; verify wire connections are firm.
- **Q: The Arduino IDE says "port not found."**
A: Ensure the USB cable is properly connected and the correct board/port is selected under Tools menu.
- **Q: All LEDs turn on at the same time.**
A: Double check pin numbers in the code match the actual wiring (pins 8, 9, 10).
- **Q: Code won't upload.**
A: Check for typos, missing semicolons, or mismatched brackets. Compare carefully with the master sketch.
- **Q: LED works but stays on permanently.**
A: Verify the correct sketch was uploaded and not an older test sketch.

9. Classroom Management Tips

- Pair confident readers with younger or less confident students to encourage peer support.
- Use color-coded stickers on wires/pins to help young learners match connections visually instead of relying on reading pin numbers.
- Keep instructions short and demonstrate each wiring step visually before students attempt it themselves.

- Build in movement breaks (like the "Red Light, Green Light" game) to manage energy levels typical of this age group.
- Have 1 teaching assistant per 2 groups for hands-on troubleshooting support.
- Celebrate small wins loudly — cheering when an LED lights up boosts engagement and confidence.

10. Follow-Up Activities

- **Take-Home Challenge:** Draw a picture of your traffic light circuit and label the wires and colors.
- **Family Activity:** Explain to a family member what each traffic light color means and why circuits need resistors.
- **Next Session Preview:** Introduce the concept of using a push-button to control when the traffic light changes (pedestrian crossing button).
- **Optional Home Practice:** If students have access to Tinkercad or similar free simulation tools, they can recreate their circuit virtually at home.

TinkerWithMe — Hands-on Arduino Robotics & Electronics Program, Nairobi